Chemistry

1A 1	2A 2	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8	8B 9	8B 10	1B 11	2B 12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01																	2 He 4.00
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.9	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (289)	116 Lv (293)	117 Ts (293)	118 Og (294)

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Water Data

T_{mp} = 0°C
T_{bp} = 100°C
c_{ice} = 2.09 J/g·K
c_{water} = 4.184 J/g·K
c_{steam} = 2.03 J/g·K
ΔH_{fus} = 334 J/g
ΔH_{vap} = 2260 J/g
K_f = 1.86 °C/*m*
K_b = 0.512 °C/*m*

Constants

R = 0.08206 L·atm/mol·K
R = 8.314 J/mol·K
R = 62.36 L·torr/mol·K
$e = 1.602 \times 10^{-19}$ C
$N_A = 6.022 \times 10^{23}$ mol^{-1}
$k = 1.38 \times 10^{-23}$ J/K
$h = 6.626 \times 10^{-34}$ J·s
$c = 3.00 \times 10^{8}$ m/s
$R_H = 2.178 \times 10^{-18}$ J
$m_e = 9.11 \times 10^{-31}$ kg
1 amp = 1 C/s
96,485 C = 1 mole of electrons = 1 $\mathcal{F}$

Activity series of metals in aqueous solution

Metal	Oxidation Reaction				
Lithium	Li	$\rightleftharpoons$	Li^+	+	e^-
Rubidium	Rb	$\rightleftharpoons$	Rb^+	+	e^-
Potassium	K	$\rightleftharpoons$	K^+	+	e^-
Barium	Ba	$\rightleftharpoons$	Ba^{2+}	+	$2e^-$
Calcium	Ca	$\rightleftharpoons$	Ca^{2+}	+	$2e^-$
Sodium	Na	$\rightleftharpoons$	Na^+	+	e^-
Magnesium	Mg	$\rightleftharpoons$	Mg^{2+}	+	$2e^-$
Aluminum	Al	$\rightleftharpoons$	Al^{3+}	+	$3e^-$
Manganese	Mn	$\rightleftharpoons$	Mn^{2+}	+	$2e^-$
Zinc	Zn	$\rightleftharpoons$	Zn^{2+}	+	$2e^-$
Chromium	Cr	$\rightleftharpoons$	Cr^{3+}	+	$3e^-$
Iron	Fe	$\rightleftharpoons$	Fe^{2+}	+	$2e^-$
Cobalt	Co	$\rightleftharpoons$	Co^{2+}	+	$2e^-$
Nickel	Ni	$\rightleftharpoons$	Ni^{2+}	+	$2e^-$
Tin	Sn	$\rightleftharpoons$	Sn^{2+}	+	$2e^-$
Lead	Pb	$\rightleftharpoons$	Pb^{2+}	+	$2e^-$
Hydrogen	H_2	$\rightleftharpoons$	$2\ H^+$	+	$2e^-$
Copper	Cu	$\rightleftharpoons$	Cu^{2+}	+	$2e^-$
Silver	Ag	$\rightleftharpoons$	Ag^+	+	e^-
Mercury	Hg	$\rightleftharpoons$	Hg^{2+}	+	$2e^-$
Platinum	Pt	$\rightleftharpoons$	Pt^{2+}	+	$2e^-$
Gold	Au	$\rightleftharpoons$	Au^{3+}	+	$3e^-$

Metals at the top of the table are most easily oxidized.